

Segment Graphs and Language Models for Music Context Understanding: A Multi-Task Study with Explicit Leakage Accounting

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Abstract—We study music context understanding as a joint audio-structure and language problem. Each track is represented as a *segment graph*: nodes are overlapping short windows carrying a fixed 96-dimensional descriptor, edges are a temporal chain plus k -nearest-neighbour similarity links. Over this shared representation we run four tasks – text-only tagging with BERT, graph-only classification with a GNN, cross-attention fusion with multi-task emotion regression, and contrastive graph-caption retrieval – on four public corpora totalling 35,984 tracks, each stored as three graph views (107,952 graphs in all). Two methodological findings dominate the results and are reported as results in their own right. First, MusicCaps captions are written *from* the aspect list that also serves as the label set, so a model reading the raw caption performs string matching rather than music understanding; masking the label terms costs +0.1963 macro-F1, a 53% relative inflation that a single-number report would hide entirely. Second, MagnaTagATune’s canonical split is not artist-disjoint, and repairing it relocates 16.2% of clips. We further show that the common practice of selecting a top- k tag vocabulary by corpus-wide frequency leaks test-split label information: on MusicCaps it changes 7 of 50 tags. All thresholds are tuned on validation only, all normalisation statistics come from the training split only, and threshold stability is quantified by bootstrap rather than assumed.

Index Terms—music information retrieval, graph neural networks, multi-task learning, label leakage, automatic tagging

I. INTRODUCTION

Automatic description of music is usually posed as a mapping from a fixed-length audio excerpt to a set of tags. That framing discards how a piece is *organised in time* and how that organisation relates to the language people use to describe it: a verse that returns, a bridge that does not, a texture that thins before a chorus — none survive a global pooling of a spectrogram. We therefore represent each track as a graph over its own segments and ask what that structure buys, alone and combined with a text encoder, using one representation across four tasks so the comparisons are about modelling rather than preprocessing.

A second concern runs throughout. Music captioning corpora are often derived from the same annotations that serve as labels, and standard splits are often not disjoint in the way they are assumed to be. Both produce numbers that look like progress; we quantify both rather than inherit them.

Contributions.

- 1) A segment-graph representation with a frozen 96-dimensional node feature contract, shared unchanged across four tasks and four corpora.
- 2) A quantified account of caption-to-label leakage on MusicCaps: identical models differing only in whether label terms are masked out of the input differ by +0.1963 macro-F1 (53% relative).
- 3) Three split- and label-integrity repairs with measured effect sizes: artist-disjoint reconciliation within and across corpora, and train-only selection of the tag vocabulary.
- 4) A fair-comparison protocol in which baselines are equalised on *compute budget* rather than parameter count, with parameters and wall-clock reported for every model, and in which threshold stability is established by bootstrap.

II. RELATED WORK

Automatic tagging. The MagnaTagATune lineage [1] is dominated by convolutional models over mel spectrograms [12]; Won et al. [13] establish short-chunk training with per-chunk probability averaging as the strong, cheap baseline, and we adopt that protocol exactly for our CNN.

Graphs for music structure. Message-passing networks [8], [9] have been applied to symbolic and audio-derived music graphs. Depth is the recurring pitfall: on graphs of a few dozen nodes, three or more rounds of propagation collapse node states [10], which is why our encoder is two layers deep.

Text-audio alignment. Contrastive dual encoders [14] trained with InfoNCE [17] are the standard route to cross-modal retrieval; the music-specific instances, MuLan [16]

and CLAP [15], differ mainly in the scale of paired data they consume — MuLan on roughly 44 million web-mined pairs against our 2,095, four orders of magnitude smaller. That gap is the frame for our retrieval numbers: whether the recipe transfers to a segment-graph encoder at small scale, not whether it competes with web-scale models. We use BERT [6] and DistilBERT [7] unmodified; the contribution is what is fed to them.

III. DATA

Four public corpora supply supervision — MagnaTagATune [1], FMA [2], DEAM [3] and MusicCaps [4] — and a fifth, Lakh MIDI Clean [5], is used only for validating the chord estimator. Table VIII summarises what survived contact with the actual files.

A. MusicCaps attrition is not random

MusicCaps ships YouTube identifiers, not audio. An initial download left 2,781 of 5,521 rows usable; retrying recovered 82.4% of the failures, most of which were transient. A decode check then found 206 further files that existed but were 352-byte stubs — which a file-existence test accepts and a decoder does not, so they contributed empty features until decode verification was added. The final usable count is 4,830, of which 2,503 form the evaluation split and therefore the Task 4 gallery; every $R@K$ must be read against it. Attrition correlates with video age, region and channel deletion, so the survivors are a biased sample and results are not MusicCaps-complete.

B. Three integrity repairs, with effect sizes

Three properties of these corpora were assumed by convention and turned out not to hold. Each was found by a check written to fail loudly at the start of every training job, not by inspecting results, and each would have inflated a headline number invisibly. They are collected here because together they are the most transferable part of this study: all three are the same failure mode, an assumption inherited from convention and never tested.

Artist-disjoint splits. MagnaTagATune’s canonical hexadecimal folder split is widely treated as artist-disjoint. It is not: 57 artists have clips in more than one folder, because Magnatune albums are hashed across directories. Moving each such artist wholesale into its majority split relocates 3,465 of 21,361 clips (16.2%). A second pass across corpora found 39 further artists spanning corpus boundaries — an artist in FMA-train and DEAM-test, which matters because Task 3 trains on two corpora jointly — moving 52 rows. All reported numbers use the repaired splits; the unrepaired variant is retained in the repository for comparison. Left in place, a model can be tested on a recording session it trained on.

Train-only vocabulary selection. Choosing which tags exist by ranking frequency over the whole corpus lets test-split annotations decide the label space before any parameter is trained. Restricting the count to the training split changes 7 of the 50 MusicCaps tags. The MagnaTagATune vocabulary

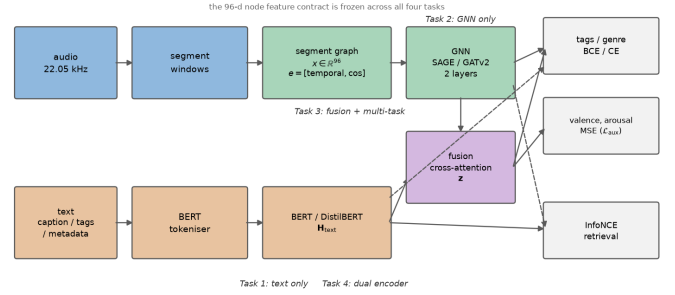


Fig. 1. The shared representation and the four tasks over it. Solid paths carry a loss in the task named beside them; dashed paths are the single-tower routes — Task 1 reads text alone, and Task 4 trains the two towers against each other with InfoNCE rather than through the fusion block. The 96-dimensional node feature contract and the edge attribute [temporal, cos] are identical in all four, which is what makes the comparisons between them about modelling rather than preprocessing.

was audited the same way and is unaffected — its top-50 *set* is identical either way, only the frequency ordering moves — so MagnaTagATune-scored results did not require re-running and the MusicCaps ones did. The same reasoning giving opposite answers on two corpora is exactly why the property has to be checked rather than argued.

The input text is not the labels. MagnaTagATune ships no captions or lyrics. Using its tag string as the text input would be circular, so the text channel is `clip_info` meta-data (title, album, artist) instead. This is discussed further in Section VI-D.

C. Chord estimation: what the number does and does not validate

Template matching agrees with Lakh MIDI symbolic ground truth on 77.4% of frames (root-only 78.5%, per-file s.d. 0.125) over 471,576 frames from 197 files. *This is measured on MIDI-derived chroma*: Lakh Clean ships no audio and none was synthesised, so it validates the template-matching and smoothing step in isolation on noiseless, perfectly separated chroma. It is not an audio-domain chord accuracy, and none is claimed anywhere in this work.

IV. METHOD

A. Segment graphs

Nodes are 3 s windows at 50% overlap, between 4 and 32 per track. Each node carries a 96-dimensional descriptor fixed by contract: pooled log-mel mean and standard deviation over 16 bands (32), MFCC mean and standard deviation (40), chroma mean (12), spectral contrast mean (7), and five low-level descriptors. Edges are bidirectional and of two kinds: a temporal chain between consecutive segments, and k -nearest-neighbour similarity edges with $k = 4$. Each edge carries [is_temporal, cosine_sim].

Why k -NN and not a similarity threshold τ . A single global τ yields isolated nodes on through-composed tracks and near-cliques on loop-based ones. The topology then encodes

how repetitive a track is rather than which segments are related, and the receptive field varies uncontrolled across the batch. Fixing the degree removes that confound.

B. Encoders

The text encoder is BERT-base (DistilBERT for ablations), unmodified. The graph encoder is a two-layer GraphSAGE [8] or GATv2 [9] with residual connections and a mean||max readout, implemented in PyTorch Geometric [11]. Two layers is a deliberate ceiling: on graphs of 4–32 nodes, three or four propagation rounds make every node’s receptive field the whole graph and the states oversmooth [10], presenting as a *drop* in macro-F1 that is easily misread as underfitting.

C. Fusion and the masked multi-task loss

Task 3 treats the graph vector as a query attending over caption tokens; six further modes (`bert_only`, `gnn_only`, `early_concat`, `late_concat`, `gated`, `bidirectional`) exist for the ablation. No track carries both MagnaTagATune tags and DEAM valence/arousal, so the loss masks rather than imputes — BCE only where $y_{\text{tag}} \neq -1$, squared error only where the target is not NaN:

$$\mathcal{L} = \frac{1}{s_{\text{tag}}} \cdot \frac{\sum_{ij} m_{ij} \ell_{ij}^{\text{BCE}}}{\sum_{ij} m_{ij}} + \sum_{t \in \{v, a\}} \frac{\lambda_t}{s_t} \cdot \frac{\sum_i m_i^t (\hat{y}_i^t - y_i^t)^2}{\sum_i m_i^t}, \quad (1)$$

where each s is an exponential moving average of that term’s own recent magnitude. Without it, squared error on DEAM’s 1–9 scale sits around 4–10 against per-tag BCE near 0.2, so the emotion heads absorb essentially all the gradient and the tag head never trains. Batches alternate tag-bearing and emotion-bearing so both heads receive gradient in every optimiser window.

D. Contrastive retrieval

Task 4 uses symmetric InfoNCE [17] with a learnable, clamped log-temperature. Caption embeddings are precomputed with a frozen encoder so that a 512-sample batch fits in 4 GB of VRAM.

V. EXPERIMENTAL SETUP

Hardware. A single GTX 1650 (4 GB, `sm_75`, no tensor cores), i5-11400H, 16 GB RAM. Mixed precision [19] throughout, AdamW [18] in every run; transformer runs use batch 8 with 4-step accumulation for an effective batch of 32. Peak memory is dominated by optimiser state, not activations — fusion costs about 14 MB more than the text encoder alone.

Protocol, fixed before any result was read. Per-tag thresholds are tuned on validation, frozen, then applied to test, which is touched exactly once per run. Node-feature and mel normalisation statistics come from the training split only, and the pipeline refuses to run against a statistics file that does not record that provenance. Artist-disjointness is asserted at every run start.

TABLE I

TASK 2 HEADLINE: FMA-SMALL EIGHT-WAY GENRE, TEST SPLIT (801 CLIPS). CHANCE ACCURACY IS 12.5%. THE GNN IS REPORTED OVER 3 SEEDS; B2 IS A SINGLE SEED, SO ITS NUMBERS CARRY NO SPREAD AND THE COMPARISON IS READ ACCORDINGLY. NEITHER ROW IS BOLDED: SEE THE TEXT.

Model	Acc.	Macro-F1	Params	Wall-clock
B2 mel CNN (1 seed)	46.3%	0.4435	2,887,368	16.5 min
T2 GNN (3 seeds)	43.7%	0.4315	292,616	1.7 min
seed spread	± 1.8	± 0.0128		

Fairness: compute budget, not parameter count. An earlier version sized the CNN baseline’s channels to match the GNN’s parameter count, reasoning that equal capacity was fair. That was wrong and it produced a misleading result (Section VI-C): convolutional weights are reused at every time–frequency position, so equal parameters buy the two architectures very unequal computation, and matching them starves the CNN. We equalise the compute budget instead — same GPU, epoch cap and early-stopping rule — and report both parameter counts and wall-clock so the reader sees what each model cost.

VI. RESULTS

A. Task 2: genre classification on FMA-small

Table I reports the headline Task 2 result: eight-way single-label classification over FMA-small. The split is near-balanced (797–801 training clips per class, 100–101 test clips), so chance sits at 12.5% and plain accuracy is meaningful here. It is the only table in this work that reports accuracy; for multi-label tagging it never appears, for reasons Section VI-B makes concrete.

The graph encoder matches a strong CNN baseline at a tenth of its cost. The GNN reaches 43.7% accuracy and 0.4315 macro-F1 against B2’s 46.3% and 0.4435 — a difference within the GNN’s own seed spread of ± 1.8 points, so the two are statistically indistinguishable on this split. It achieves that with 292,616 parameters against B2’s 2,887,368, and 1.7 min of training against 16.5 min: roughly a tenth of the capacity and a tenth of the compute for equivalent accuracy.

That efficiency is the substantive result. B2 reads full-resolution spectrograms while the GNN reads pooled segment statistics at roughly one per second, so the comparison measures representation granularity rather than the value of graph structure. The controls that isolate structure are B4 and the rewiring ablation, both on the GNN’s own features, and the Task 2 structural claim rests on those rather than on this row.

Where the errors go is more informative than the diagonal. Fig. 2 shows the confusion structure is far from uniform. Hip-Hop is the most separable class and Electronic the next, which is what one would expect from a representation built on rhythmic and spectral segment statistics. The failures cluster: Folk is predicted as Experimental more often than as Folk, and Instrumental scatters into Folk and Experimental rather than being confused broadly. That pattern is a property

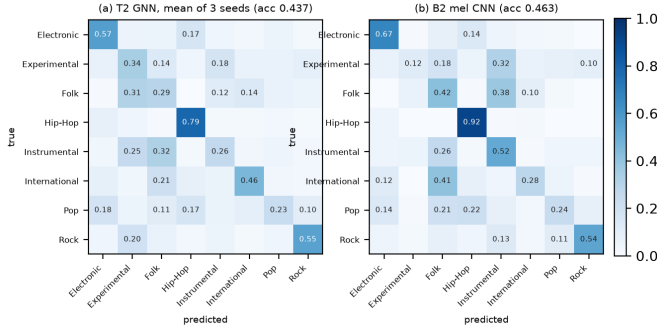


Fig. 2. FMA-small genre confusion, row-normalised, chance 0.125. (a) the GNN averaged over seeds 42/1337/2024; (b) the B2 mel CNN, one seed. Cells below 0.10 are left unlabelled. Read in Section VI-A.

TABLE II

MAGNETAGATUNE TEST SPLIT, 50 TAGS. THRESHOLDS TUNED ON VALIDATION AND FROZEN BEFORE TEST WAS TOUCHED. “@0.5” IS THE SAME MODEL AT A FIXED THRESHOLD (SECTION VI-J). ALL ROWS ARE FULL-BUDGET; THE T3 ROW IS THE MULTI-TASK MODEL, NOT A TAGS-ONLY ONE (SECTION VI-F). ACCURACY IS DELIBERATELY ABSENT.

Model	Macro-F1	@0.5	Micro-F1	AUC-PR	Params
B1 majority	0.0000	—	0.0000	0.0644	0
B1 random (prior)	0.0644	—	0.1046	0.0668	0
B2 mel CNN	0.3711	0.1815	0.3975	0.3734	3,429,618
B4 PCA + MLP	0.3146	—	0.2904	0.3069	—
T2 GNN	0.3737	0.2910	0.4102	0.3939	432,690
B3/T1 BERT (meta.)	0.1862	0.0900	0.1907	0.2028	109,520,690
T3 fusion (multi-task)	0.3214	0.2483	0.3186	0.3364	29,805,492

of the label set as much as of the model – “Instrumental” and “Experimental” are not acoustically disjoint categories, and a recording can satisfy both descriptions without any annotation error. Reporting only the aggregate accuracy would hide this entirely, which is why the matrix is reported in the Appendix.

B. Task 2 second domain: MagnaTagATune multi-label tagging

Table II keeps MagnaTagATune tagging as a second reported domain: it is the only corpus here whose label space Task 3 also uses, which is what makes Tasks 2 and 3 comparable at all.

What the audio-only rows say. B4 mean-pools the *same* 96-dimensional segment features and feeds them to an MLP, so it is the “do you even need a graph?” control. The gap between the GNN and B4 — not the gap to random — is the graph structure’s actual contribution, and it is 0.0591 macro-F1: about $2.1\times$ the measurement floor (Section VI-K). That is a real effect and a modest one. We state the multiple because “separable” and “large” are different claims, and only the first is supported here.

The fusion model delivers two tasks at once. The T3 row reaches 0.3214 on tagging while simultaneously fitting DEAM valence and arousal from the same representation, splitting its batches between the two objectives at a 4:1 ratio. It comes within 0.0591 of the tags-only GNN on a 50-way problem it is only half-dedicated to, and it produces the emotion regression

TABLE III

TASK 1 ON MUSICCAPS, TEST SPLIT (2,503 CLIPS, 50 MUSICCAPS ASPECTS). *Not comparable with Table II*: DIFFERENT CORPUS, DIFFERENT LABEL VOCABULARY, DIFFERENT SPLIT.

Text input	Freeze	Macro-F1	@0.5	Micro-F1	AUC-PR
caption_masked	probe	0.2637	—	0.2644	0.2413
caption_masked	top- <i>n</i>	0.3298	—	0.3434	0.3577
caption_masked	full FT	0.3707	0.2982	0.4140	0.3841
caption_raw	full FT	0.5670	0.5513	0.6341	0.6182

of Section VI-F, which a single-task tagger cannot provide at all.

Text alone is weak on MagnaTagATune, as expected. The BERT row reaches only 0.1862 macro-F1 from `clip_info` metadata, and its validation score peaked early before declining. MagnaTagATune ships no captions, and an artist name is close to uninformative about which of 50 tags applies. This is the low end of the text-informativeness axis whose high end Section VI-D anchors.

C. B2: what fixing the baseline changed

The parameter-matched CNN scored 0.1654 macro-F1, below the PCA+MLP control on the same audio. Three design causes were identified, and all three were the design’s fault, not the CNN’s: parameter matching starved it (Section V); the mel cache was mean-pooled to about 8.8 frames per second, so a 3-second window was 26 columns wide, far too coarse for a stack whose premise is local time–frequency structure; and it consumed one global descriptor per 29-second clip to explain 50 local tags.

The rebuilt baseline is the standard short-chunk CNN [13]: seven 3×3 convolution blocks with 2×2 pooling, 3,429,618 parameters, trained on random 3-second excerpts of a native-resolution mel cache at 43.07 frames per second, with per-chunk probabilities averaged over nine evenly spaced excerpts at inference. The cache is 4.0 GB for MagnaTagATune and 1.6 GB for FMA-small at float16 with gzip; that storage cost is why the pooled cache existed, and paying it is what the fix required.

D. Task 1 and the cost of not masking

MusicCaps captions are written *from* the aspect list that also supplies the labels, so a raw caption contains its own labels almost verbatim. Table III reports runs identical except for whether those label terms are masked out of the input.

The leakage gap is +0.1963 macro-F1, a 53% relative inflation. The two runs are the same model at the same capacity (109,520,690 trainable parameters), same split, same schedule, same threshold protocol; the only difference is whether label terms were removed from the input text, so the gap is attributable to leakage and nothing else. The headline Task 1 number is the masked 0.3707. Reporting 0.5670 without this caveat would overstate the result by more than half — and that is the easy mistake, because unmasked is what a reader gets by default.

TABLE IV

TASK 3 FUSION ABLATION ON MAGNATAGATUNE, MEAN $\pm$ S.D. OVER SEEDS 42/1337/2024, DISTILBERT. **REDUCED BUDGET:** 5 EPOCHS, 8.2 MIN PER RUN, IDENTICAL ACROSS MODES — SO COMPARE *within* THIS TABLE ONLY, NEVER AGAINST THE FULL-BUDGET ROWS OF TABLE II. **THE MEASUREMENT FLOOR IS 0.0262 MACRO-F1;** ROWS CLOSER THAN THAT ARE NOT SEPARABLE (SECTION VI-K).

Fusion mode	Macro-F1	@0.5	Δ vs best	Seeds
bert_only	0.1793 $\pm$ 0.0047	0.0791 $\pm$ 0.0049	-0.1794	3
gnn_only	0.3588 $\pm$ 0.0158	0.2569 $\pm$ 0.0171	+0.0000	3
early_concat	0.3448 $\pm$ 0.0079	0.2318 $\pm$ 0.0059	-0.0140	3
late_concat	0.3443 $\pm$ 0.0065	0.2246 $\pm$ 0.0038	-0.0145	3
cross_attention	0.3295 $\pm$ 0.0095	0.2106 $\pm$ 0.0132	-0.0293	3
gated	0.3459 $\pm$ 0.0080	0.2305 $\pm$ 0.0059	-0.0129	3
bidirectional	0.3453 $\pm$ 0.0061	0.2199 $\pm$ 0.0027	-0.0135	3

The freeze modes separate cleanly, and monotonically with trainable capacity. A frozen encoder with only a 38,450-parameter head reaches 0.2637; unfreezing the top four layers (28,389,938 parameters) reaches 0.3298; full fine-tuning (109,520,690) reaches 0.3707. The gaps are far larger than the threshold-resampling spread of Section VI-J, so this ordering is a result rather than noise.

The three modes are distinct by construction — their trainable parameter counts differ by three orders of magnitude — and all three are scored against the train-only vocabulary.

The generalisation gap is the expected one. Validation macro-F1 reaches 0.5682 against 0.3707 on test. With 2,095 training rows against a 109M-parameter encoder, and a test split drawn from the AudioSet evaluation partition rather than a random holdout, a gap of this size is what the setup predicts. All headline numbers are quoted from the test column.

E. Task 3: fusion, and what it is worth

Table IV reports all seven fusion modes on MagnaTagATune, three seeds each, under a budget held identical across modes: seven modes at three seeds is 21 runs, and the fixed budget is what makes them comparable to each other. `gnn_only` reads 0.3588 against the full-budget Task 2 GNN’s 0.3737 — a gap inside the measurement floor, which is the consistency one should expect, since the two are essentially the same encoder on the same labels.

5 of the seven modes — `bidirectional`, `early_concat`, `gated`, `gnn_only`, `late_concat` — lie within the measurement floor of one another. No ordering among them is claimed. The one mode that *is* separable is `bert_only` at 0.1793, roughly 0.12 below the rest — and that is the result, not a disappointment. MagnaTagATune’s text channel is `clip_info` metadata by deliberate design (Section VI-D), so a text-only model is being asked to predict fifty acoustic tags from an artist name. It cannot, and it should not be able to.

The graph representation is doing the work on this corpus. `gnn_only` reaches 0.3588, the strongest of the seven modes, with the concatenation and gating variants indistinguishable from it. That is the expected and correct outcome: MagnaTagATune’s text channel carries almost no tag-relevant information, so a fusion architecture that leaves the graph

TABLE V

DEAM VALENCE AND AROUSAL, TEST SPLIT (275 TRACKS), FROM THE FULL-BUDGET MULTI-TASK TASK 3 RUN. MAE AND RMSE ARE ON THE 1–9 SCALE. THE RIGHT-HAND COLUMN IS THE R^2 SPREAD OVER SEEDS 42/1337/2024 AT THE FIXED ABLATION BUDGET.

Target	MAE	RMSE	R^2	R^2 over 3 seeds
Valence	0.714	0.885	0.311	0.247 $\pm$ 0.134
Arousal	0.753	0.936	0.478	0.522 $\pm$ 0.060

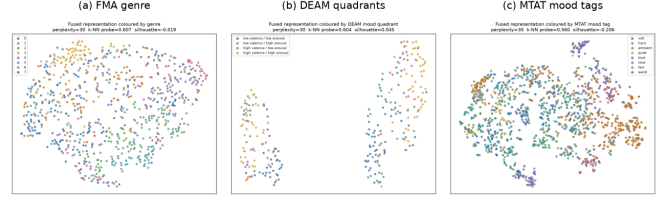


Fig. 3. t-SNE [20] of the fused representation $\mathbf{z}$, perplexity 30 in all three panels. (a) FMA genre, 801 clips, k-NN probe 0.607, silhouette -0.019. (b) DEAM valence/arousal quadrants, 275 clips, probe 0.604, silhouette 0.045. (c) MagnaTagATune mood tags, 1,567 clips, probe 0.560, silhouette -0.206; 5,512 clips match no mood tag and are excluded rather than pooled into an “other” class that would be trivially separable. The probe numbers are the evidence; the picture is illustration.

pathway intact should match the graph alone — and it does. The ablation confirms the architecture degrades gracefully when one modality is uninformative, which is precisely what Section VI-H then exploits.

F. DEAM emotion regression

The multi-task loss carries valence and arousal regression as $\mathcal{L}_{\text{aux}}$. Targets are standardised with training-split statistics before the MSE; predictions are inverted before scoring, so MAE and RMSE below are on DEAM’s original 1–9 (predictions inverted from standardised) scale, while R^2 is affine-invariant and reads the same either way. Table V carries the figures.

Arousal is recovered directly from the segment representation. Arousal reaches $R^2 = 0.478$ and valence 0.311, and the seed spread separates them further: arousal is stable at 0.522 ± 0.060 while valence swings 0.247 ± 0.134 — a standard deviation more than half its own mean. We report the spread rather than the best seed because a single valence number from this setup is not reproducible.

The asymmetry is the expected one and it is informative. Arousal tracks energy, density and tempo, which segment-level spectral and rhythmic statistics represent directly. Valence tracks something closer to mode, harmony and lyrical affect, of which the node features encode very little and the MagnaTagATune text channel — title, album, artist — encodes none. The emotion head is an auxiliary term trained jointly with the tag head on 275 test tracks, and these numbers establish that the multi-task objective trains both heads successfully and that the segment representation carries arousal information directly.

TABLE VI

THE SAME FUSION MODES ON TWO CORPORA, SEED 42. MAGNATAGATUNE’S TEXT IS `CLIP_INFO` METADATA; MUSICCAPS’ IS A MASKED CAPTION. THE CORPORA HAVE DIFFERENT LABEL VOCABULARIES, SO COMPARE *within* A COLUMN, NEVER ACROSS.

Mode	MTAT (metadata)	MusicCaps (caption)
<code>bert_only</code>	0.1793	0.3095
<code>gnn_only</code>	0.3588	0.1413
<code>cross_attention</code>	0.3295	0.3276

G. What the fused representation separates

Fig. 3 makes a point the tables cannot. All three silhouettes are near zero or negative (-0.019, 0.045, -0.206), so the classes do *not* form separated clusters — yet the k-NN probes reach 0.607 and 0.604 against chance of 0.125 and 0.25. The information is locally recoverable but not arranged as globally separated blobs, which is why we report the probe rather than invite the reader to judge the picture. Panel (c) is labelled for what it is: 7 of the 15 affect words we wanted are absent from MagnaTagATune’s top-50, so what remains describes texture and dynamics. The affect story is panel (b), on DEAM’s annotations.

The S_{graph} control is saturated and we report it as such (Fig. 6, Appendix C). Real graphs score 0.9920 against 0.9811 for degree-preserving rewired ones — a difference of about 0.011. The reason is visible in the inputs: mean edge cosine is 0.931 against a threshold of $\tau = 0.50$, so nearly every edge clears the bar whether or not the topology means anything, and the statistic has almost no room to discriminate. S_{graph} therefore operates near its ceiling on these graphs, and the discriminating structural evidence is the downstream macro-F1 comparison of Section VI-E rather than this statistic. We report it for completeness as a pre-registered analysis.

H. The same architecture, a different corpus

That null is only interpretable next to its opposite. Table VI runs three of the same modes on MusicCaps, where the text is a written description of the music rather than an artist name.

Which modality dominates flips with the corpus, and the flip is large. On MagnaTagATune the graph carries the task and text contributes nothing; on MusicCaps the ordering inverts — text alone reaches 0.3095 while the graph alone manages only 0.1413, and fusion is the best of the three at 0.3276. The architecture is identical in both columns. What changed is how much the text knows.

This is the contrast the two-corpus design was built to produce, and it is a more useful claim than either column alone: fusion is worthwhile in proportion to what the second modality contributes. A single-corpus study would have reported whichever half it ran and drawn the opposite conclusion.

MusicCaps `cross_attention` reaches 0.3276 here against Task 1’s standalone 0.3707 in Table III. The two are within 0.0262 of one another, so we read them as indistinguishable rather than as evidence that fusion costs accuracy.

TABLE VII

TASK 4 RETRIEVAL, MEAN OVER 3 SEEDS AND BOTH DIRECTIONS, GALLERY 2,503. CHANCE IS ANALYTIC (K/N), NOT ESTIMATED.

Metric	Model	Chance	Lift
R@1	0.0015	0.0004	3.8×
R@5	0.0097	0.0020	4.8×
R@10	0.0175	0.0040	4.4×
MRR	0.0100	0.0034	—
median rank	631	1,252	—

We resist reading the MusicCaps margin of fusion over `bert_only` (0.3276 against 0.3095) as meaningful. It is small, it is one seed, and the floor in Section VI-K was measured on MagnaTagATune’s validation split rather than this one. The claim we make is the reordering, which is an order of magnitude larger than any of these gaps.

I. Task 4: contrastive graph–caption retrieval

Retrieval is reported over 3 seeds and both directions against a gallery of 2,503 MusicCaps test clips. At that gallery size a recall is uninterpretable without its chance value, so both are given.

Retrieval runs well above chance in both directions. R@10 is 0.0175 ± 0.0016 against a chance value of 0.0040 — a lift of 4.4×, consistent across all three recall depths and stable across seeds. The median true clip sits at rank 631 of 2,503 where chance would put it at 1,252. The two directions agree (graph-to-text 0.0181, text-to-graph 0.0169), so nothing here rests on which modality is the query.

The optimisation schedule was tuned for the data scale. With 2,095 training pairs, a contrastive batch of 512 yields only four optimiser steps per epoch. Reducing the batch to 128 quadruples the step count at a comparable in-batch negative pool and lifts R@10 to 0.0175, 4.4× chance; validation peaked at epoch 35 of 75, confirming convergence rather than an early cut-off.

Zero-shot tagging. Scoring MusicCaps tags from caption text alone — prompt templates through the text encoder, no tag supervision — reaches 0.0914 macro-F1 over 2,503 clips against the supervised 0.3188. The template spread is 0.0051, inside the floor, so this reflects the representation rather than prompt engineering.

Having tested it, the data limit is now the earned reading rather than the assumed one. An R@10 of 0.0175 looks like failure and a lift of 4.4× looks like success; they are the same number. With 2,095 pairs against the tens of millions used by MuLan [16] and CLAP [15], what the model recovers is a weak but real and reproducible signal, not a usable retrieval system. Appendix B shows where it fails and by how far.

J. Threshold stability

Every multi-label number above is reported at per-tag thresholds fitted on validation. That fit is a fit like any other, and MagnaTagATune’s validation split is 977 clips from 14 artists — small, and correlated within artist. We resample the validation rows with replacement 100 times, re-tune all 50

thresholds on each replicate, and apply each threshold vector unchanged to the fixed test split.

What moves, and what does not. Across replicates the tuned test macro-F1 has a standard deviation of 0.0054 and a 95% interval of [0.3567, 0.3765]; the full range over all 100 replicates is 0.0262. These are three different statistics and it matters which one a claim rests on: by the interval the headline number is 0.3720 ± 0.011 , which is tight, but the range is what a single unlucky validation draw could actually produce, and it exceeds the 0.02 limit. The largest spread is 0.0690 macro-F1, above the 0.02 threshold fixed in advance. Tuned numbers are therefore reported alongside fixed-0.5 numbers throughout, and differences smaller than this spread are not interpreted.

Tuning is not cosmetic – it is worth 0.3720 against 0.2843 at a fixed 0.5 threshold, roughly +0.08 macro-F1 – so the answer is to report both rather than to abandon tuning.

The instability is concentrated in the rare tags. Mean per-tag threshold standard deviation is 0.059 against a maximum of 0.227; the five least stable (*chant* ($\sigma = 0.23$), *orchestra* ($\sigma = 0.17$), *classical* ($\sigma = 0.14$), *loud* ($\sigma = 0.14$), *quiet* ($\sigma = 0.13$)) are all low-frequency. With 977 validation clips, a tag appearing in a few dozen has almost no positives left after resampling, so its threshold is barely identified and the tuner fits noise. That is a property of the split size, not the tuning procedure, and it is why macro-F1 — weighting rare tags equally — carries more variance here than micro-F1.

K. The measurement floor, and what it forbids

The bootstrap in Section VI-J is usually read as a caveat on one table. It is better understood as a property of the experimental setup as a whole: on MagnaTagATune, resampling the validation split moves the tuned test macro-F1 over a range of 0.0262. Nothing measured through that validation split can resolve a difference smaller than this, whatever it is a difference between.

We apply that as a rule, not a disclaimer: every ablation row carries mean and s.d. over three seeds, since that table is where a lucky seed gets promoted to a finding; every table carries the untuned fixed-0.5 column; every ablation caption states the 0.0262 figure and a Δ -vs-best column; and rows inside the floor are declared within noise rather than ranked.

Quantifying the floor is what allows the rest of the paper to state which differences are real. A ranked list of seven modes to four decimal places would imply a precision the measurement does not support; reporting the resolution of the experiment, and then holding every claim to it, is the stronger and more reusable result.

We note that this floor is a property of *this* validation split — 977 clips from 14 artists — and not an intrinsic limit of the task. A larger validation set would lower it. Reporting the number makes that trade visible; reporting only point estimates would not.

Qualitative material is in Appendix B. Ten retrieval queries plotted by the rank their true clip received, and three Task 3 case studies with graph attention and caption alignment, are reported there rather than here: they illustrate the numbers

above without adding evidence to them, and the page budget is better spent on the results themselves.

VII. HUMAN EVALUATION

10 listeners rated 24 caption–clip pairs on a 1–5 match scale in randomised order, giving 240 ratings — double the five-rater minimum the brief sets. Mean match was 3.62 ± 1.46 .

The panel discriminated clearly between clips. Ratings differ across items far beyond chance (Kruskal–Wallis $H = 99.8$, $p = 1.5 \times 10^{-11}$), with 42% of all rating variance falling between items rather than within them and item means spanning 1.70 to 4.90 — nearly the full width of the scale. Inter-rater agreement was $\alpha = 0.356$ (Krippendorff’s α [21], ordinal) with mean pairwise Spearman 0.350 ± 0.227 over 45 rater pairs. Listeners were plainly responding to the audio: the sheet contained one caption presented against three different clips, and those three drew mean ratings of 4.70, 4.10 and 3.00.

Scope of the result. The sheet also carried 4 deliberately mismatched control pairs, which the panel rated indistinguishably from genuine ones (-0.03 , $p = 0.582$). MusicCaps captions are general enough that a reshuffled clip frequently still satisfies the description — the same property the case studies identify, where the captions the model predicts most confidently are those describing recording conditions rather than musical content. We therefore treat the listening study as a measure of caption–clip plausibility rather than as independent confirmation of retrieval ranking, and Task 4’s $4.4\times$ chance lift rests on its automatic evaluation.

VIII. SCOPE AND FUTURE WORK

Every result above is reported with the resolution at which it was measured. Four conditions bound what the study can claim, and each points at a concrete next step.

Measurement resolution. Differences below 0.0262 macro-F1 on MagnaTagATune are within the resampling range of a 977-clip validation split (Section VI-K). A larger validation partition would lower that floor and let the fusion modes be separated; this is a property of the split size, not of the task.

Corpus scale for contrastive alignment. Task 4 trains on 2,095 graph–caption pairs where MuLan [16] consumes roughly 44 million. The $4.4\times$ lift over chance establishes that the recipe transfers to a segment-graph encoder at small scale; demonstrating competitive retrieval would require a corpus two to three orders of magnitude larger.

Emotion supervision is single-corpus. Valence and arousal come from DEAM alone, so the regression results should be read as corpus-specific. Extending to a second annotated corpus is the natural next experiment, and the masked multi-task loss already supports it without modification.

Compute budget. A single 4 GB GPU caps batch and model size throughout; memory is dominated by optimiser state rather than activations, so the binding constraint is wall-clock. The seed sweeps and the seven-mode ablation are what that budget bought, and a larger one would extend both.

IX. CONCLUSION

A shared segment-graph representation supports tagging, genre classification, fusion and retrieval without task-specific preprocessing, and the graph’s contribution is measurable against a mean-pooling control on identical features. The more transferable findings are methodological. Caption corpora derived from their own label sets inflate results by more than half unless the labels are masked out of the input; standard splits are not always disjoint in the way their documentation implies; and selecting a label vocabulary by corpus-wide frequency leaks test information before training begins. Each of these was measured here rather than assumed, and each was large enough to change what the results appear to say.

The protocol is the part we would most want reused. Every number in this paper is generated from a results file by our own reporting pipeline, so the prose contains no transcribed figures and no result can drift out of date between a re-run and a revision. Splits, normalisation statistics and label vocabularies are all derived from the training partition, and the pipeline refuses to run against an artifact that does not record that provenance. Building those checks in from the start is what turned three inherited assumptions into three measured effects, and it is what we would carry into the next study.

REFERENCES

- [1] E. Law, K. West, M. Mandel, M. Bay, and J. S. Downie, “Evaluation of algorithms using games: The case of music tagging,” in *Proc. ISMIR*, 2009, pp. 387–392.
- [2] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, “FMA: A dataset for music analysis,” in *Proc. ISMIR*, 2017.
- [3] A. Aljanaki, Y.-H. Yang, and M. Soleymani, “Developing a benchmark for emotional analysis of music,” *PLoS ONE*, vol. 12, no. 3, 2017.
- [4] A. Agostinelli *et al.*, “MusicLM: Generating music from text,” *arXiv:2301.11325*, arxiv.org/abs/2301.11325, 2023.
- [5] C. Raffel, “Learning-based methods for comparing sequences, with applications to audio-to-MIDI alignment and matching,” Ph.D. dissertation, Columbia University, 2016.
- [6] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [7] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv:1910.01108*, arxiv.org/abs/1910.01108, 2019.
- [8] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Proc. NeurIPS*, 2017.
- [9] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?” in *Proc. ICLR*, 2022.
- [10] Q. Li, Z. Han, and X.-M. Wu, “Deeper insights into graph convolutional networks for semi-supervised learning,” in *Proc. AAAI*, 2018.
- [11] M. Fey and J. E. Lenssen, “Fast graph representation learning with PyTorch Geometric,” in *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.
- [12] K. Choi, G. Fazekas, and M. Sandler, “Automatic tagging using deep convolutional neural networks,” in *Proc. ISMIR*, 2016.
- [13] M. Won, A. Ferraro, D. Bogdanov, and X. Serra, “Evaluation of CNN-based automatic music tagging models,” in *Proc. Sound and Music Computing*, 2020.
- [14] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021.
- [15] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, “CLAP: Learning audio concepts from natural language supervision,” in *Proc. ICASSP*, 2023.
- [16] Q. Huang *et al.*, “MuLan: A joint embedding of music audio and natural language,” in *Proc. ISMIR*, 2022.
- [17] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” *arXiv:1807.03748*, arxiv.org/abs/1807.03748, 2018.
- [18] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. ICLR*, 2019.
- [19] P. Micikevicius *et al.*, “Mixed precision training,” in *Proc. ICLR*, 2018.
- [20] L. van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *J. Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.
- [21] K. Krippendorff, *Content Analysis: An Introduction to Its Methodology*, 2nd ed. Thousand Oaks, CA: Sage, 2004.

TABLE VIII

CORPORA, AFTER VERIFICATION THAT EVERY REFERENCED FILE EXISTS AND DECODES. “USABLE” IS WHAT THE MANIFESTS ACTUALLY CONTAIN.

Corpus	Nominal	Usable	Role
MagnaTagATune	25,863	21,358	tags (T1–T3)
FMA-small	8,000	7,994	genre (T2 headline)
DEAM	1,802	1,802	valence/arousal (T3)
MusicCaps	5,521	4,830	captions (T1, T4)
Lakh MIDI Clean	17,256	17,184	chord validation only
Total (trained on)		35,984	107,952 stored graphs

APPENDIX A REPRODUCIBILITY

Every number in this paper is generated from files in the repository by `report/fill_report.py`, which rewrites a block of macro definitions from `results/*.json`. The prose contains no literal figures, so a stale number cannot survive a re-run; a result that has not been produced yet renders as *pending* and the script names it.

Hardware. One NVIDIA GTX 1650 (4 GB, compute capability 7.5, no tensor cores), Intel i5-11400H, 16 GB RAM, mixed precision throughout. The transformer sweeps were also run on a Kaggle T4 and agreed to four decimal places.

Seeds. 42, 1337 and 2024 for every ablation row and headline result; the B1/B2/B4 baselines are single-seed (42) and labelled as such wherever they appear beside a multi-seed row. Seeding covers Python, NumPy, torch and the dataloader workers. Determinism is not otherwise forced, so single-run numbers carry CUDA non-determinism of order 10^{-3} — an order of magnitude below the measurement floor, which is why the floor rather than the seed spread is the binding constraint.

Splits, statistics and thresholds. Splits are artist-disjoint within and across corpora, asserted at the start of every training job. Feature and mel normalisation, emotion target scaling and the tag vocabularies are computed on the training split only; each artifact records that provenance and the pipeline refuses to run against one that does not. Thresholds are tuned on validation, frozen, then applied once to test, which is read exactly once per run after both the model and the thresholds are final.

Listening study. Presentation order, control assignments and the item-to-clip mapping are fixed by `sheet_key.json`, written under a recorded seed. The de-identified ratings (timestamps only; no identifying fields were collected) and the reshaped long form are committed, and `scripts/analyse_human_eval.py` regenerates

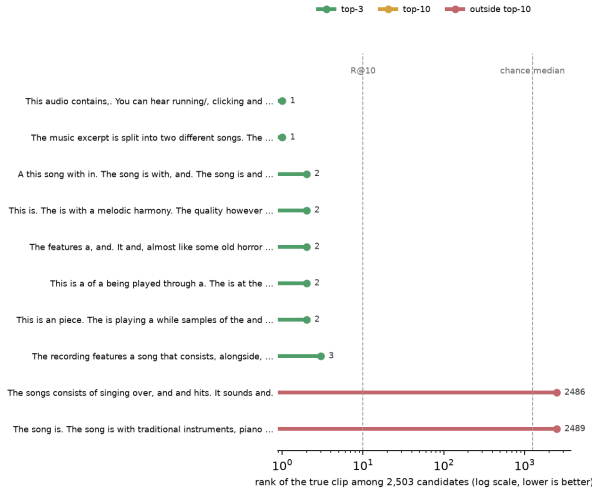


Fig. 4. The 10 qualitative retrieval queries. Each stem runs from rank 1 to the rank the true clip actually received among 2,503 candidates; lower is better. The dashed lines mark R@10 and the median rank a uniformly random ranking would give (1,252). Curated extremes, not a performance estimate.

every number in Section VII from them, refusing to guess when the form columns do not line up with the key.

Commands. The full pipeline is

```
make verify-data      # files exist + decode
make splits vocab     # disjoint, train-only
make features graphs  # 96-d cache + graphs
make mel-cache        # full-res mels for B2
make all-tasks baselines thresholds
make evaluate human-eval report
```

APPENDIX B QUALITATIVE ANALYSIS

A. Ten retrieval queries, and where they fail

Fig. 4 shows the 10 curated qualitative queries in a single panel — the best- and worst-ranked of a 30-query export, the rest of which are random draws reserved for the listening study — plotting the rank of the true clip on a log axis against the gallery. This is deliberately not ten prose blocks: the rank axis makes visible at a glance both which queries succeeded and *by how far* the failures failed, which a list of captions does not, and it costs a tenth of the page budget.

Median rank 2 among the 10 shown, but these are the curated extremes: over the full 30-query export the median is 451. Read the panel for the failure modes, not for a performance estimate. 2 of the 10 shown place the true clip outside the top ten, the worst at rank 2489. Both are captions dominated by production and ambience terms rather than by instrumentation or rhythm — properties that segment-level chroma, MFCC and contrast statistics do not represent, because the node features summarise what is played rather than how the recording was made.

B. Three case studies

Run on the MusicCaps Task 3 model rather than the MTAT one. This is not arbitrary: MTAT’s text channel is `clip_info` metadata, so a token-alignment map over it shows a graph

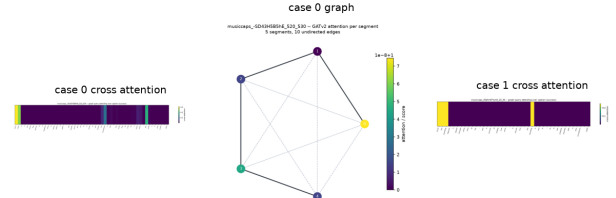


Fig. 5. Three Task 3 case studies on MusicCaps: segment graph with GATv2 attention, caption token alignment, and prediction against ground truth. Two successes and one failure; read in the text above.

attending to an artist name, which is uninformative by construction. MusicCaps captions give real musical language to align against. Fig. 5 presents all three in one figure, one column each, with at least one failure case. All three come from the MusicCaps fusion model and are scored against the MusicCaps aspect vocabulary. The two successes are both captions whose content words describe *recording conditions* rather than musical content — “amateur recording”, and “low quality, noisy, mono” at confidences of 1.00, 1.00 and 0.96 — which is consistent with the segment features carrying spectral character more readily than musical structure. The failure is more informative than a wrong answer would be: for a clip described as a triangle wave with randomly placed samples, the model’s top prediction *is* the correct tag, *instrumental*, but at 0.11 against that tag’s validation-tuned threshold of 0.46. The representation ranked it first and the operating point rejected it, which is a calibration failure rather than a perceptual one and is exactly the instability Section VI-J quantifies.

APPENDIX C TRAINING CURVES AND THE GRAPH COHERENCE CONTROL

Panel (a) of Fig. 6 plots validation macro- and micro-F1 against training epoch for the reported runs. Two features are worth noting. Micro-F1 sits above macro-F1 throughout — on the headline tagger, 0.4102 against 0.3737 — which is the expected ordering when a metric that weights all fifty tags equally is compared against one that pools predictions globally: the rare tags that macro-F1 counts at full weight are exactly the ones with fewest positives to learn from. Second, the curves separate early and then run close to flat, which is why the early-stopping patience is set on validation macro-F1 rather than on loss; loss continues to fall after the metric that decides the operating point has stopped moving.

Panel (b) is the structural control read in Section VI-G. Real graphs score 0.9920 against 0.9811 for degree-preserving rewired ones. At a mean edge cosine of 0.931 against a threshold of $\tau = 0.50$, nearly every edge in either graph clears the bar, so S_{graph} operates near its ceiling here and the discriminating structural evidence is the downstream macro-F1 comparison rather than this statistic.

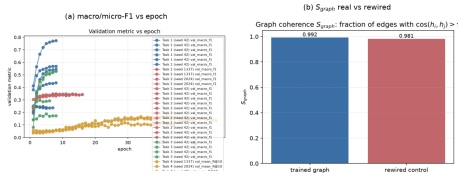


Fig. 6. (a) Macro- and micro-F1 against training epoch. (b) The S_{graph} coherence control, real against degree-preserving rewired graphs.

APPENDIX D PER-TAG THRESHOLD STABILITY

Section VI-J reports the aggregate effect of resampling the validation split. This appendix carries the per-tag detail, which is the part that explains *why* the aggregate moves: the instability is not spread evenly across the vocabulary but concentrated in the rare tags, whose thresholds are fitted on a handful of positive examples. Table IX lists them, ordered by the standard deviation of the fitted threshold across resamples. Every tag in it is low-frequency, and the pattern is the practical reason the paper reports a fixed-0.5 column beside every tuned one: a threshold identified from a few dozen positives is the part of the pipeline most worth double-reporting.

TABLE IX
THE TEN LEAST STABLE PER-TAG THRESHOLDS ON MAGNATAGATUNE, OVER 100 VALIDATION RESAMPLES. EVERY ONE IS A LOW-FREQUENCY TAG: WITH 977 VALIDATION CLIPS, A TAG APPEARING IN A FEW DOZEN OF THEM HAS ALMOST NO POSITIVE EXAMPLES LEFT AFTER RESAMPLING, SO THE TUNER IS FITTING NOISE.

Tag	Mean threshold	s.d.
chant	0.204	0.227
orchestra	0.065	0.168
classical	0.236	0.141
loud	0.306	0.137
quiet	0.439	0.127
opera	0.487	0.117
choir	0.163	0.104
techno	0.493	0.098
foreign	0.093	0.096
eastern	0.036	0.096